Family OS

Comprehensive End-to-End Workflow Specification

VERSION 0.1.0 | FGP DETERMINISTIC GOVERNANCE | REAL MARKETPLACE ADB AGENT | HARD PAYMENT BOUNDARY

Family OS -- How It Works: The Visual Workflow Guide

A Simple, Step-by-Step Visual Guide from Initial Setup to Safe Autonomous Checkout

Target Audience: Everyone (Family Members, Developers, and Evaluators)
Core Story: Asking for a **"Keyboard under Rs. 3,000"** and watching the phone safely navigate to checkout.

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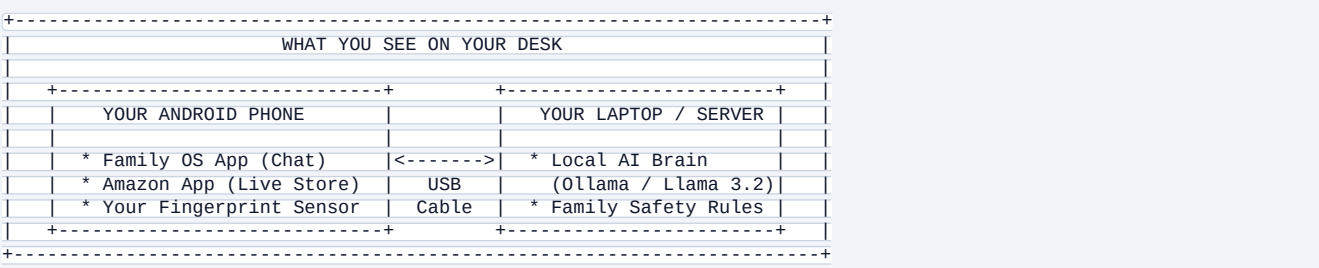
1. The 30-Second Big Picture

Family OS is a private, family-safe AI assistant that helps your family find products and autonomously prepares them for purchase on your physical Android phone -- **without ever spending your money without your permission.**

Unlike ordinary chatbots that suggest fake products or generic smart speakers that buy things without asking, Family OS connects your laptop's private AI brain directly to your physical smartphone via USB:



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2. The 3 Golden Rules of Family OS

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|                        THE 3 GOLDEN RULES                             |
|                                                                       |
|  1. ZERO FAKE DATA (Real Marketplace)                                 |
|     The assistant never invents products. It opens real Amazon on     |
|     your phone, checks live prices in Rupees, and reads real stock.   |
|                                                                       |
|  2. DETERMINISTIC SAFETY (The Family Bouncer)                         |
|     The AI never decides if someone is allowed to buy something.      |
|     Strict family rules govern spend limits, roles, and approvals.    |
|                                                                       |
|  3. THE HARD PAYMENT BOUNDARY (You Keep Control)                      |
|     The assistant drives the phone through search, cart, and address, |
|     but STOPS DEAD at the payment screen. It leaves Amazon open so    |
|     YOU authorize the final payment with your fingerprint or UPI PIN. |
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3. Master Workflow Diagram

Here is the entire journey from your spoken words to the final checkout screen:

FLOWCHART LOGIC OVERVIEW

A1. You Speak or Type 'Find keyboard under 3000' -> B2. Safety Check FGP Is this member allowed to shop?

B -- ALLOW -> C3. Private AI Brain Extracts: Keyboard, Max Rs. 3,000

B -- DENY -> B1Polite Stop: Permission Denied

C -> D4. Phone Takes Action Opens Amazon on phone & reads live listings

D -> E5. Smart Scoring Ranks by Price, Prime Delivery & Reviews

E -> F6. Recommendation Cards Shows top choices in Family OS app

F -> G7. You Tap 'Proceed to Checkout' Picks Logitech K380 at Rs. 2,499

G -> H8. Approval Needed? e.g., Child Account

H -- YES Child -> H1Parent Notification Admin approves in Governance tab

H1 -> I9. Autonomous Phone Takeover Agent taps item, adds to cart, confirms address

H -- NO Adult -> I

I -> J10. HARD PAYMENT BOUNDARY Agent halts touches at payment screen!

J -> K11. You Finish Payment Touch fingerprint sensor or enter UPI PIN

4. Step-by-Step Walkthrough: "Keyboard under Rs. 3,000"

Let's follow one complete, real-life scenario from start to finish.

Step 1: Connecting the System (Plug & Play)

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|                        HARDWARE CONNECTION                              |
|                                                                         |
|   1. Plug phone into laptop with USB cable (enable USB Debugging).      |
|   2. The laptop runs a quick port-forwarding command (adb reverse).     |
|   3. Open the Family OS app on your phone.                              |
|                                                                         |
|   Result: Phone and laptop are in instant, private communication.       |
|   Zero Wi-Fi firewall issues. Zero cloud eavesdropping.                 |
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- **On your laptop:** The private AI brain (Ollama) and the Family OS server run locally.
- **On your phone:** You open the Family OS mobile app.

Step 2: You Ask for a Product

You tap the microphone or type in the Purchase Agent chat:
"Find keyboard under 3000"

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|  WHAT HAPPENS IN 0.1 SECONDS:                                           |
|                                                                         |
|  * If you typed: The text is sent straight to the assistant.           |
|  * If you spoke: The app records your voice while holding the button,  |
|    converts it to text, and IMMEDIATELY DELETES the audio recording.   |
|    No raw voice files are ever stored. Privacy is 100% guaranteed.     |
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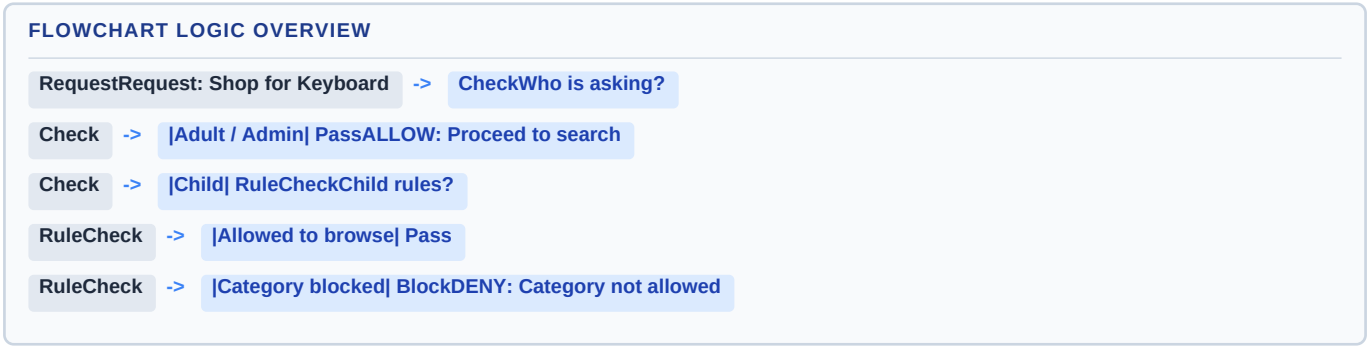
Step 3: The Family Bouncer Checks Permissions

Before the AI even looks at your message, the **Family Governance Platform (FGP)** checks your house rules:



- **Adults & Parents:** Can search freely and checkout up to their family limit.
- **Children:** Can search and explore, but **cannot purchase** without parent approval.
- **Deterministic:** The AI model never decides permissions. Rules are written in strict code.

Step 4: The Local AI Brain Understands Your Intent

Your request is sent to the local AI model (Llama 3.2) running on your laptop:

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|                        AI INTENT PARSING                                |
|                                                                         |
|   Input:   "Find keyboard under 3000"                                   |
|                                                                         |
|   AI Figures Out:                                                       |
|   * Category:  KEYBOARD                                                 |
|   * Search:    keyboard                                                 |
|   * Budget:    Rs. 3,000 (INR)                                          |
|   * Action:    Ready to search live marketplace immediately             |
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The AI does not stall with useless questions. It immediately triggers live product discovery.

Step 5: Real Shopping on Your Phone (Live Amazon Scraping)

Now watch your phone! The assistant uses **Android Debug Bridge (ADB)** to inspect the live marketplace:

FLOWCHART LOGIC OVERVIEW

Start1. Assistant wakes up phone display -> **OpenAmazon**2. Opens Amazon India to 'keyboard under 3000'

OpenAmazon -> **ReadScreen**3. Reads product titles & live prices from screen

ReadScreen -> **Scroll**4. Automatically swipes down to see more choices

Scroll -> **ReadMore**5. Reads page 2 listings

ReadMore -> **ReturnApp**6. Brings Family OS app back to your screen

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|  WHY THIS IS DIFFERENT FROM CHATGPT / ALEXA:                            |
|                                                                         |
|  * Alexa suggests sponsored items linked to one default account.        |
|  * ChatGPT invents prices and links that are often outdated or broken.  |
|  * Family OS reads genuine, live prices on your actual phone screen     |
|    right now, filtering out any item above Rs. 3,000.                   |
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Step 6: Smart Multi-Factor Ranking & Product Cards

The assistant grades every keyboard it found using an unbiased scoring formula:

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|                        HOW CANDIDATES ARE SCORED                        |
|                                                                         |
|   + 15 pts: Base score for any valid product                            |
|   + 40 pts: Price is within your Rs. 3,000 budget                       |
|   + 20 pts: Brand match (Logitech, HP, Redragon, Portronics)            |
|   + 15 pts: Currently in stock with fast delivery                       |
|   + 10 pts: Full warranty & 7-day replacement details available         |
|                                                                         |
|   TOTAL: Up to 100 points                                               |
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What You See on Your Phone Screen:

The Family OS app displays a clean carousel of product cards:

Rank	Product Title	Real Price	Delivery	Action
#1	**Logitech K380 Multi-Device Wireless**	**Rs. 2,499**	Free One-Day Prime	`[Proceed to Checkout ->]`
#2	**Redragon K552 Mechanical Keyboard**	**Rs. 2,799**	Two-Day Delivery	`[Proceed to Checkout ->]`
#3	**HP 150 Wireless Keyboard & Mouse**	**Rs. 1,299**	One-Day Prime	`[Proceed to Checkout ->]`
#4	**Portronics Bubble Multimedia Wireless**	**Rs. 999**	One-Day Prime	`[Proceed to Checkout ->]`

Step 7: You Select & Proceed to Checkout

You pick the **Logitech K380 at Rs. 2,499** and tap **"Proceed to Checkout ->"**.

FLOWCHART LOGIC OVERVIEW

TapYou tap 'Proceed to Checkout' -> **CheckRoleIs user an Adult or Child?**

CheckRole -> **|Adult Within Limit| AutoApprovedApproved Immediately Proceeds to phone automation**

CheckRole -> **|Child Account| NeedsApprovalHALT: Parent Approval Required Notification sent to Admin**

NeedsApproval -> **ParentDecidesParent reviews on their phone**

ParentDecides -> **|Approved| AutoApproved**

ParentDecides -> **|Rejected| DeniedTask Cancelled politely**

Step 8: Autonomous Phone Takeover (Touch by Touch)

Once approved, the **Mobile-Use Agent** drives the Amazon app on your phone. You can watch your phone screen as it performs each action:

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|                     WATCH YOUR PHONE SCREEN!                            |
|                                                                         |
|   Step 1: Opens Amazon directly to the Logitech K380 page.              |
|   Step 2: Locates and taps the Logitech K380 product card.              |
|   Step 3: Taps the yellow "Buy Now" button (or "Add to Cart").          |
|   Step 4: Confirms your saved home delivery address.                    |
|   Step 5: Detects any Amazon Prime upsell popups ("Join Prime for       |
|           Rs. 1499") and automatically taps "No Thanks".                |
|   Step 6: Lands on the final Payment Screen (UPI / Cards / GPay).       |
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Step 9: The Hard Payment Boundary (Safety Stop!)

This is the most critical security design in Family OS:

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|                                                                       |
|                    STOP! THE HARD PAYMENT BOUNDARY                    |
|                                                                       |
|   The phone is now on the Amazon Payment Screen showing:              |
|   * Payment options: UPI (Google Pay, PhonePe), NetBanking, Cards     |
|   * Final Order Total: Rs. 2,499.00                                   |
|   * Delivery Address: Your Home                                       |
|                                                                       |
|   ================================================================    |
|                  THE ASSISTANT STOPS TOUCHES HERE                     |
|   ================================================================    |
|                                                                       |
|   The AI will NEVER:                                                  |
|   X Click "Pay Now" or "Place Your Order"                             |
|   X Enter your UPI PIN                                                |
|   X Enter your card CVV or bank OTP                                   |
|   X Read your biometric sensors                                       |
|                                                                       |
|   WHAT YOU DO:                                                        |
|   1. Pick up your phone.                                              |
|   2. Review the Rs. 2,499 total on Amazon.                            |
|   3. Touch your fingerprint sensor or enter your UPI PIN to pay.      |
|                                                                       |
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5. Summary Table: What You See vs. What Happens Behind the Scenes

Stage	What You Experience on Phone	What Happens Behind the Scenes	Safety Guarantee
1. Request	Speak or type: *"Find keyboard under 3000"*	Audio converted to text; raw audio file deleted immediately	No voice recordings stored
2. Policy Check	Quick instant spinner	Deterministic FGP engine verifies your family role and spend limit	AI cannot bypass permissions
3. Intent	*"Searching for keyboards under Rs. 3,000..."*	Local Llama 3.2 model parses category and budget limit	100% private on your laptop
4. Discovery	Phone opens Amazon, scrolls, and returns to app	Assistant dumps screen accessibility nodes; maps prices to items	Zero fake or outdated data
5. Selection	You see ranked cards; tap **Proceed to Checkout**	Multi-factor scorer ranks options (`15` base + `40` budget + `15` stock)	Unbiased, mathematical ranking
6. Approval	Adult proceeds; child sees "Awaiting parent approval"	Parent reviews request in Governance console and approves	Kids cannot spend money
7. Navigation	Watch phone tap Buy Now, pick address, and skip upsells	Grounded coordinate taps via ADB bridge	Fully transparent touch trace
8. Checkout	Phone screen pauses on UPI / Payment options	**Touch execution halts dead.** Agent marks task complete	**Human fingerprint / PIN required**

6. Frequently Asked Questions

Can the assistant ever spend my money without me knowing?

No. Never. The Hard Payment Boundary is built into the engine code. The agent is physically programmed to halt before any payment button. You must physically authorize the purchase using your fingerprint, face ID, or UPI PIN.

What if my phone is not connected via USB?

If no phone is plugged in, Family OS automatically switches to its built-in development catalogue so you can still test and explore without errors.

Do I need to pay for an expensive cloud AI subscription?

No. The AI runs 100% locally on your laptop using free, open-source models via Ollama (Llama 3.2). Your family data never leaves your home network.

Family OS -- Safe, private, and autonomous home assistance with complete human control.